

THEOREM. *In an integral domain $\mathcal{A}$, for every polynomial P in the variable x with coefficients in $\mathcal{A}$, $P \in \mathcal{A}[x]$, and every $r \in \mathcal{A}$, there exist unique $Q \in \mathcal{A}[x]$ and $R \in \mathcal{A}$ such that*

$$P(x) = (x - r)Q(x) + R,$$

where

$$R = P(r).$$

LEMMA. *From any difference of two n -th powers, $x^n - c^n$ (where $n \geq 1$), $(x - c)$ can be factored out. That is, there exists a polynomial $Q_{n-1}(x)$ such that**

$$x^n - c^n = (x - c)Q_{n-1}(x).$$

○ (by mathematical induction on n)

The **base case** $n = 1$ holds since

$$x^1 - c^1 = (x - c)(1),$$

where the constant $1 = Q_0(x)$ is a polynomial as well.

Inductive Hypothesis. Assume for some arbitrary positive integer $k \geq 1$ there exists a polynomial $Q_{k-1}(x)$ such that

$$x^k - c^k = (x - c)Q_{k-1}(x).$$

Induction Step. To show that the lemma holds for $k + 1$ (assuming it holds for k), consider

$$x^{k+1} - c^{k+1},$$

that can be rewritten by adding $0 = -xc^k + c^kx$ and grouping as

$$x^{k+1} - xc^k + c^kx - c^{k+1} = (x^{k+1} - xc^k) + (c^kx - c^{k+1}).$$

Factoring out the common components from each group gives

$$x(x^k - c^k) + c^k(x - c).$$

* Explicitly, $Q_{n-1}(x) = \sum_{k=0}^{n-1} x^{n-1-k} c^k = x^{n-1} + x^{n-2}c + \cdots + xc^{n-2} + c^{n-1}$

Applying the inductive hypothesis to substitute $(x - c)Q_{k-1}(x)$ for $(x^k - c^k)$ yields

$$x(x - c)Q_{k-1}(x) + c^k(x - c).$$

Factoring out the common multiplier $(x - c)$ results in

$$(x - c)(xQ_{k-1}(x) + c^k),$$

where $Q_k(x) = xQ_{k-1}(x) + c^k$ is a polynomial*.

This completes the induction step, proving the lemma for all $n \geq 1$. ●

Let $P_n(x)$ be a polynomial in $\mathcal{A}[x]$ of degree n ,

$$P_n(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0,$$

where each coefficient $a_i \in \mathcal{A}$.

* Since $Q_{k-1}(x)$ is a polynomial by the inductive hypothesis, and polynomials are closed under multiplication by x and addition of constants, $Q_k(x) = xQ_{k-1}(x) + c^k$ is also a polynomial.